

Improving Preparedness of Communities for Evacuations using ZEVs

Project Overview

Meeting #1 with Community Partners

Ricardo de Castro (UC Merced)

April 26th, 2024

Agenda

Time	Description	Presenter
11:00- 11:30	Round of Introductions (1min each)	-
11:30-11:45	Project Overview	Ricardo de Casto (UCM)
11:45-12:05	Readiness for EV-based evacuation	Keith Corzine, Osman Saleem (UCSC)
12:05-12:30	Planning Infrastructure Investments	Keith Corzine, Saheed Lakeh (UCSC)
12:30-13:25	Lunch	-
13:25-13:30	Group Photo	-
13:30-14:00	Minimizing Evacuation Time via Optimization Tools	Xuchang Tang (UCD), Shuang Feng (UCM)
14:00-14:20	Definition of use cases for Merced/Mariposa evacuations	Qijian Gan(UCB)
14:20:14:40	Potential Public Policy Implications for effective EV-based evacuations	S. Markolf (UCM)
14:40-15:00	Conclusion & Next Steps	

*10 min presenting + 10 min Q&A

Round of Introductions

1. Name?
2. Affiliation?
3. Experience with evacuations?
(professional or personal)
4. What is your expectation for this meeting?

#	Organization	Name
1	UC Merced	Ricardo de Castro
2	UC Merced	Sam Markolf
3	UC Merced	Shuang Feng
4	UC Merced	Dilawer Ali
5	UC Merced	Sarah Kurtz
6	UC Merced	Farzan ZareAfifi
7	UC Davis	X. Lin
8	UC Davis	<u>Xuchang Tang</u>
9	UC Santa Cruz	Keith Corzine
10	UC Santa Cruz	Osman Saleem
11	UC Santa Cruz	Saeed <u>Lakeh</u>
12	<i>UC Berkeley</i>	Scott Moura
13	UC Berkeley	<u>Qijian Gan</u>
14	<u>Calstart</u>	Valerie Thorsen
15	<i>Merced Office of Emergency Services</i>	Adam Amaral
16	<i>Merced Fire Chief</i>	Derek Parker
17	<i>SJ Air Valley Air Pollution District</i>	Kevin Wing
18	SJ EV Partnership	Ollie Danner
19	Mariposa Office of Emergency Services	Wes
20	Perimeter	Noah Wu
21	UC Merced Office of Emergency	Laura Mascorro

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Natural Hazards Trigger Evacuations in California

2023: Planada (Merced) Enters Third Day of Evacuations
as More **Storms** Loom Ahead ...



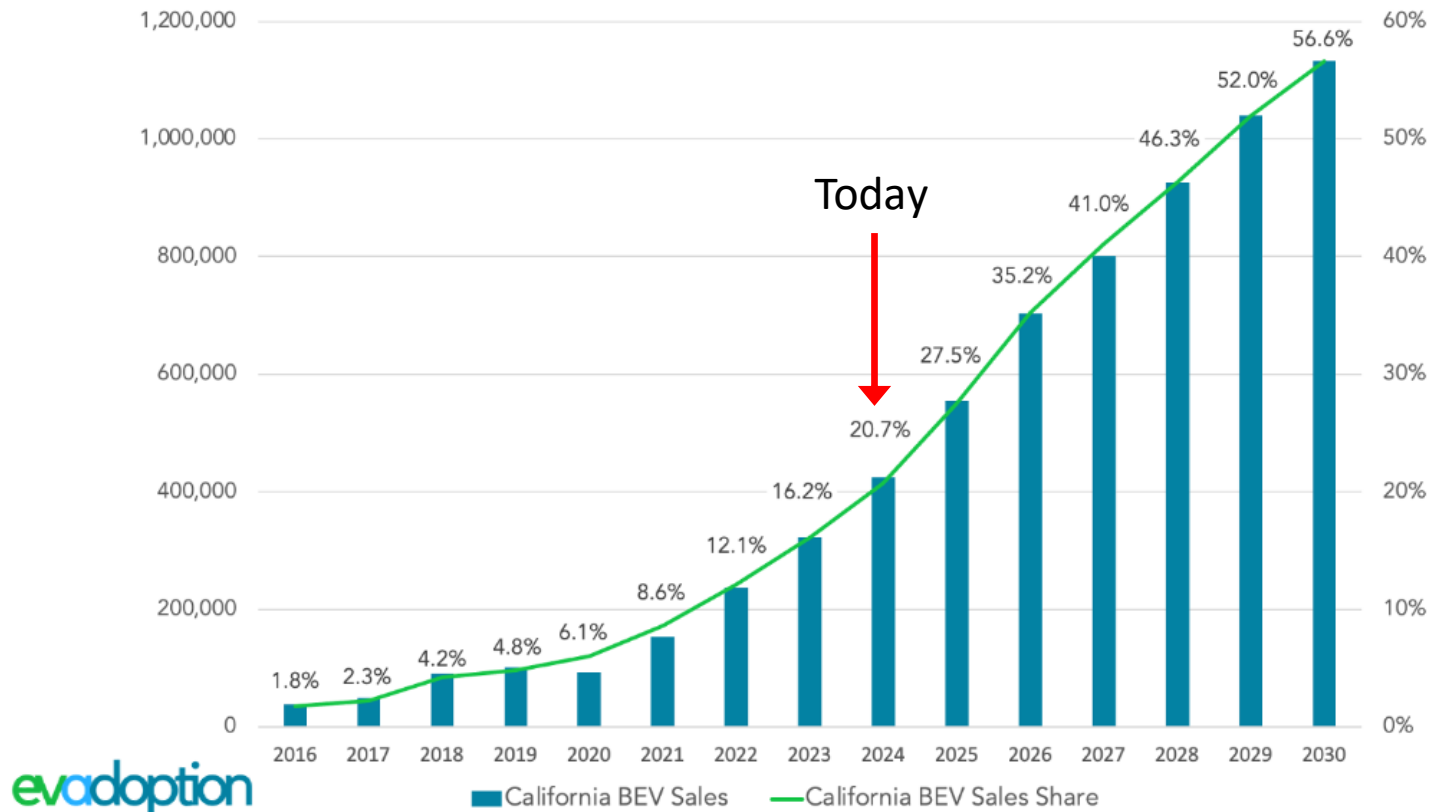
2022: Oak **fire** near Yosemite explodes to 6555
acres. Thousands evacuated



2007: mass evacuations (**500,000**
people) over a 7day period in San
Diego due to **Witch Fire**

Mass adoption of EVs in California

California BEV Sales & New Vehicle Share Forecast: 2016-2030



CA Executive order N-79-20:
100% sales of light duty Zero
Emission Vehicles (ZEV) by
2035

**Q: Are we ready to deal with
evacuations using only EVs?
What could be the challenges?**

Challenge 1: Lack of Power

California faces historic power outage due to fire danger

BY ASSOCIATED PRESS

OCTOBER 8, 2019

Share   

SAN FRANCISCO (AP) — Millions of people were poised to lose electricity throughout northern and central California after Pacific Gas & Electric Co. announced Tuesday it would shut off power in the largest preventive outage in state history to try to avert wildfires caused by faulty lines.

1. How to evacuate when **grid power is not available** to charge EVs?

Challenge: Grid Overload

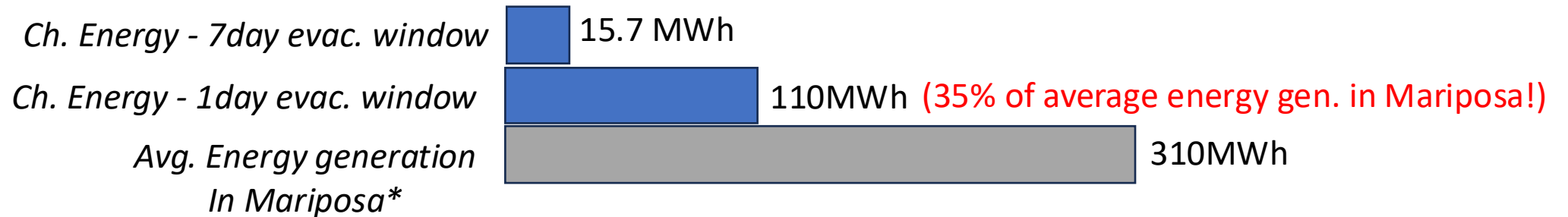
Assumptions

- Evacuation of 5k cars in Mariposa
- 0.32kWh/mile of energy consumption
- 70 miles to reach “safety”/shelter

Evacuation Energy Needs

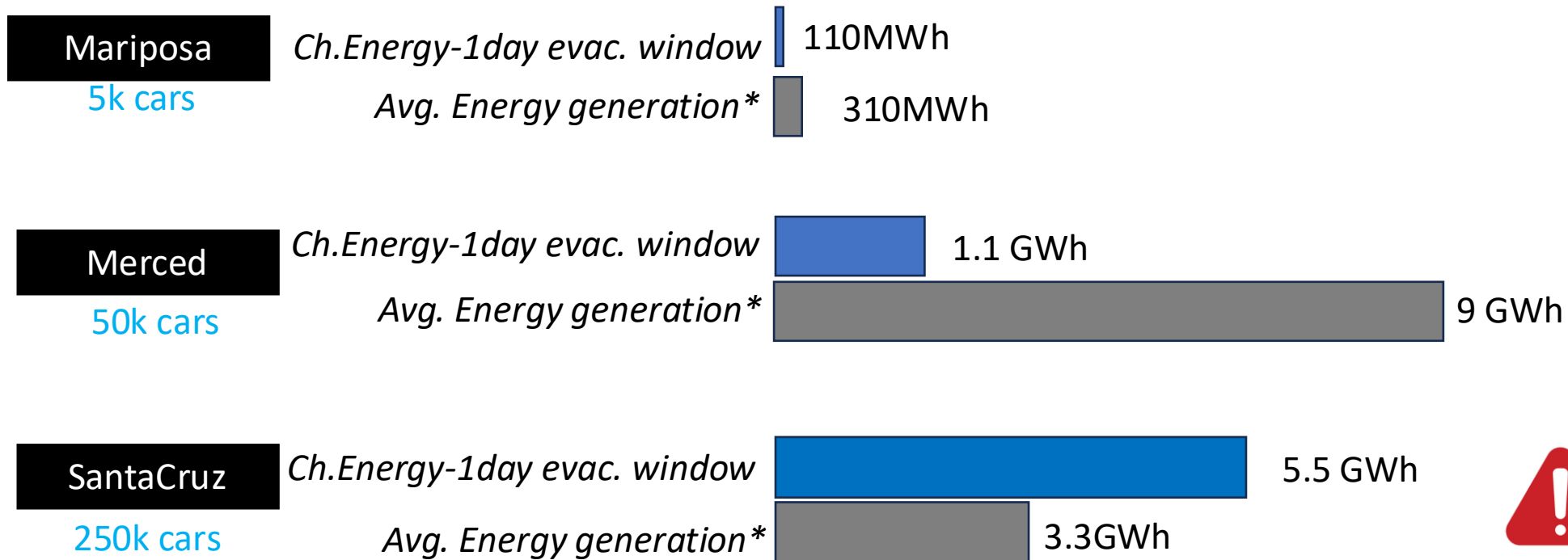
- Energy: 22kWh/vehicle
- Total Evac. Energy: 22kWh*5k = **110MWh**

Short evacuation window may **overload the grid!**



*Mariposa avg power in 2022: 114GWh/365=0.31GWh/day=**13MW**

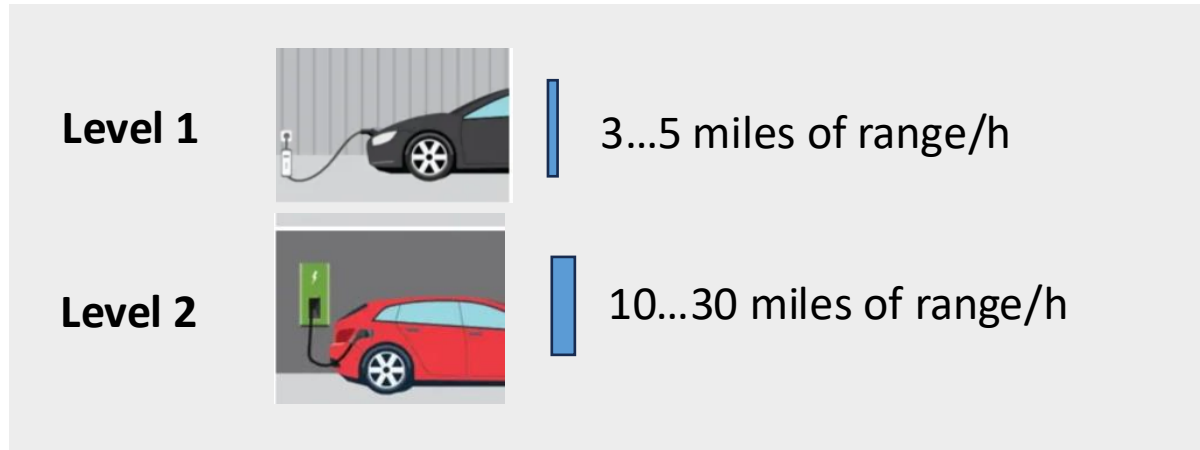
Challenge: Grid Overload



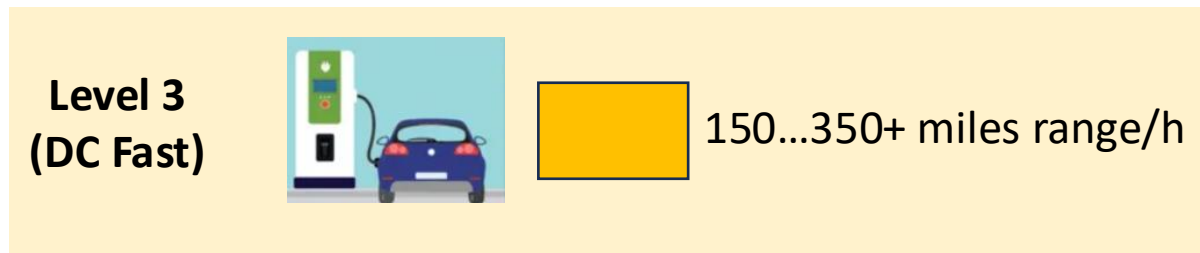

Ch Energy Demand > Avg Energy Generation!

*What if we need to evacuate and the grid is **overloaded**?*

Challenge: Charging Time



→ L1/L2 charge at home is good option if we have long time to evacuate



→ L3 Charging more attractive for “quick” evacuations, but still 10x to 30x slower than gasoline vehicles



3: How to minimize charging time/queuing during evacuation?

Credits: City of Phoenix

Challenge: Public Policy

- “When backup power becomes law” (Fla. Stat. § 526.143)
 - *requires gas stations located along evacuation routes to have backup power to pump fuel so residents can evacuate before a storm*
 - *business must keep a written statement attesting to the periodic testing and ensured operational capacity of the equipment.*
 - *required documents must be made available, upon request, to the county emergency management office*

4: Should we mandate Backup power in EV charging stations that are located along critical evacuation routes?

Source: Hurricane Irma’s Effect on Florida’s Fuel Distribution System and Recommended Improvements

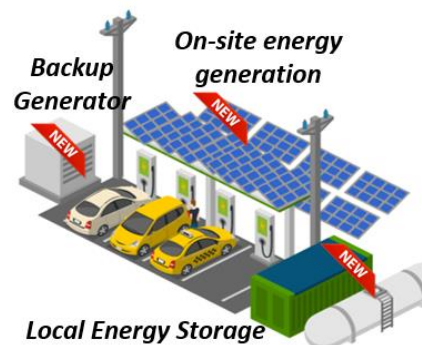
Project Outcomes/Objectives

O1. A new ZEV Evacuation Readiness Score

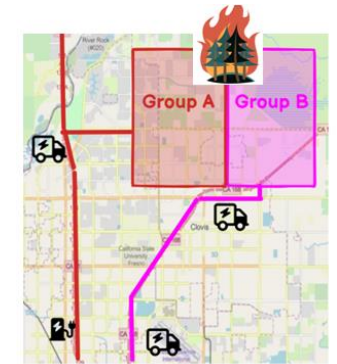


O2/O3: Decision Aid Tools

O2. planning investments and reduce vulnerabilities



O3. Evacuation plans (route, charging scheduling)



O4. Policy Recommendations



White paper to support effective ZEV evacuations

O5. Resources/Education Materials/Outreach

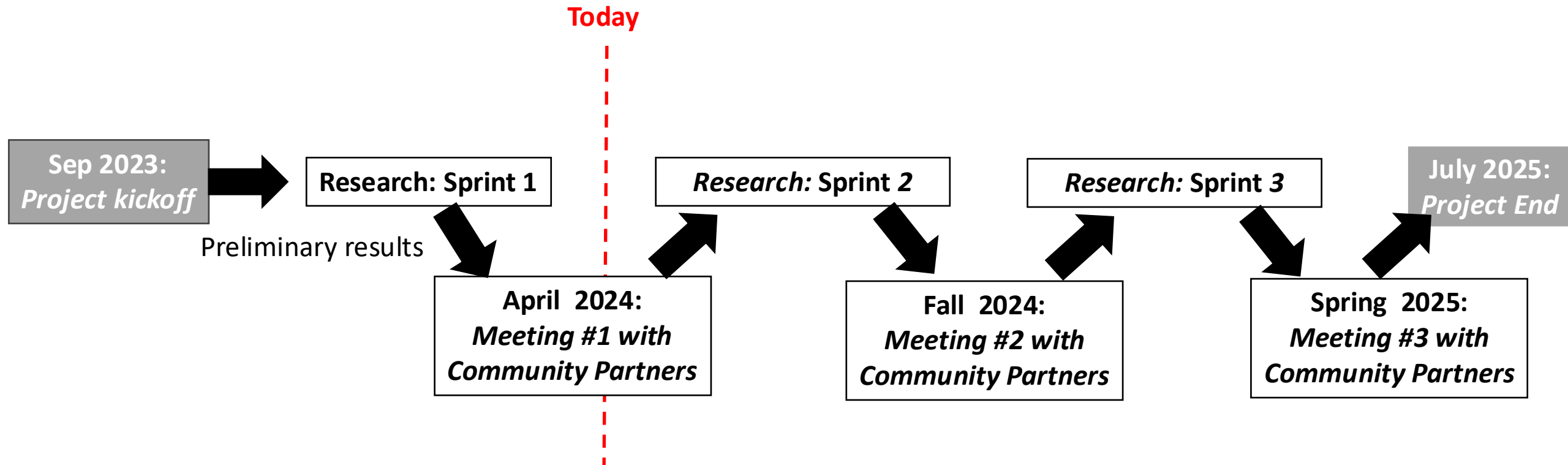


Factsheets, checklists, flowcharts, workshops

Team & Community Partners



Project Timeline



Share with community partners progress of the research team

Objectives for the Day

- Share with community partners progress of the research team
- Collect feedback from partners. Some questions:
 - Do you think the goals are sound and have practical relevance?
 - Are you aware of other EV challenges during evacuations that we overlooked?
 - We would like to build evacuation use cases for Central Valley/Sierra Nevada/Sacramento/Bay Area/Santa Cruz. Are you available to help us create these use cases?

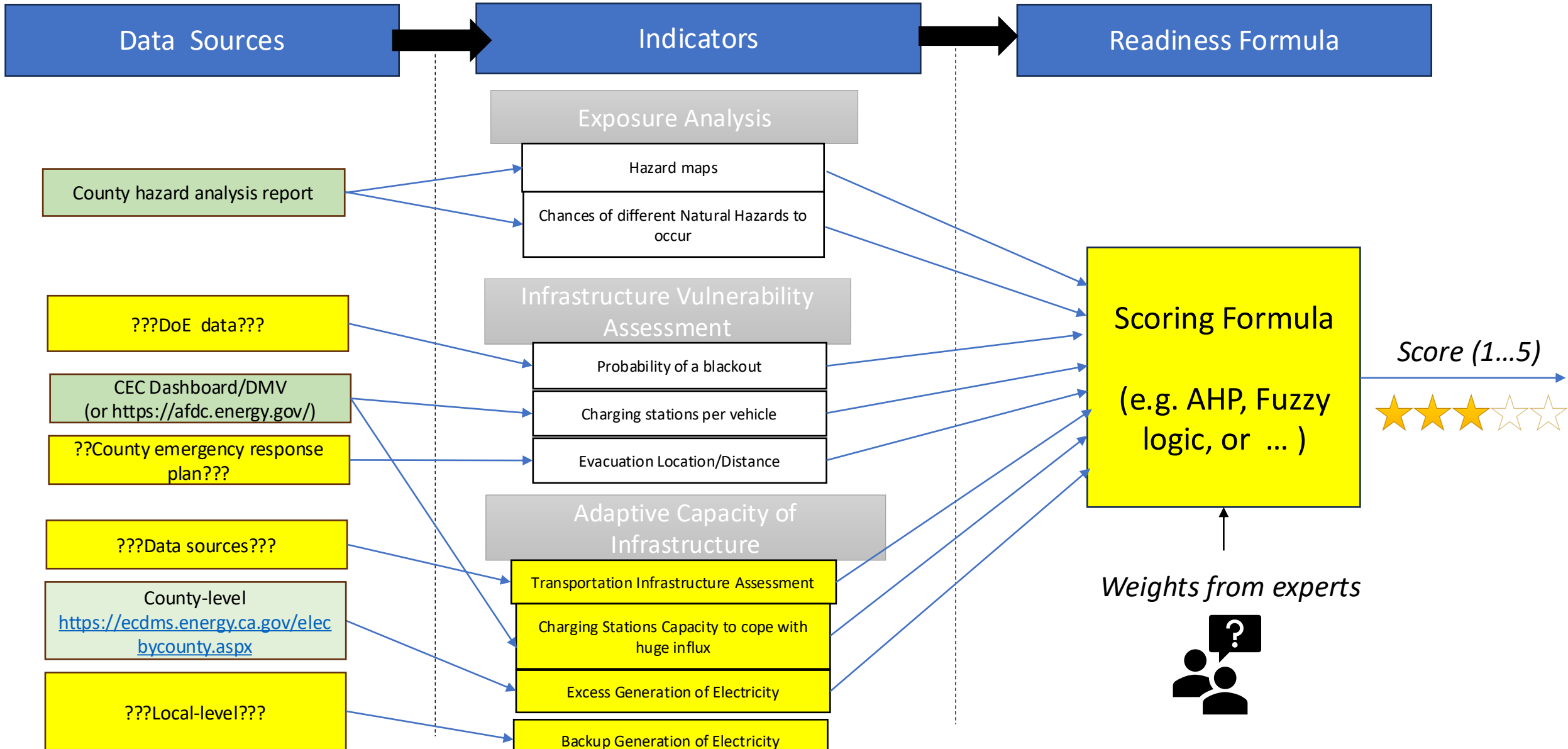
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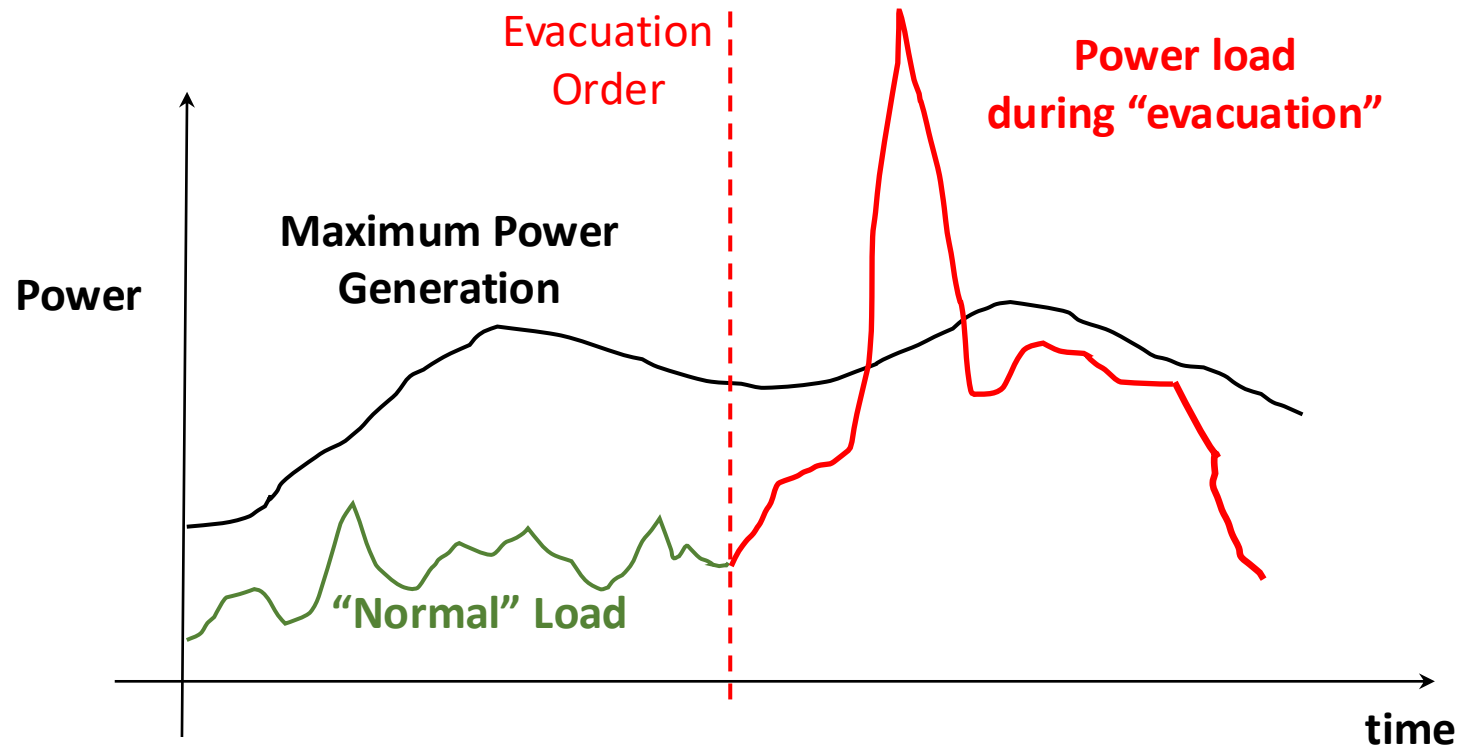
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- Extra Slides

Readiness Score: Overview



Challenge 2: Grid Overloading



2. What if we need to evacuate and the grid is **overloaded**?

Challenge: Grid Overload

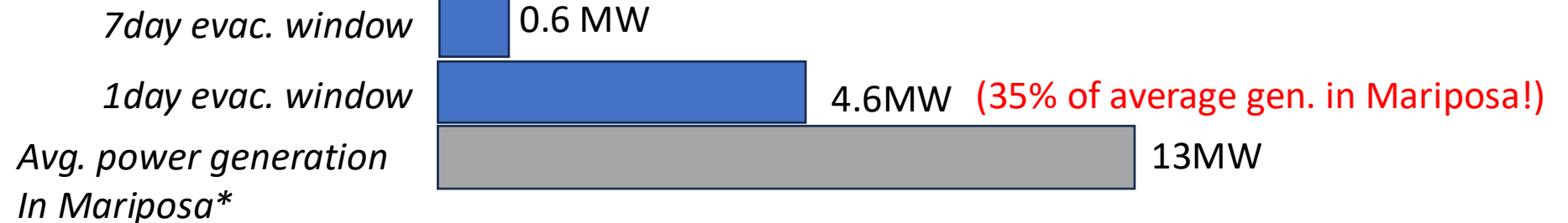
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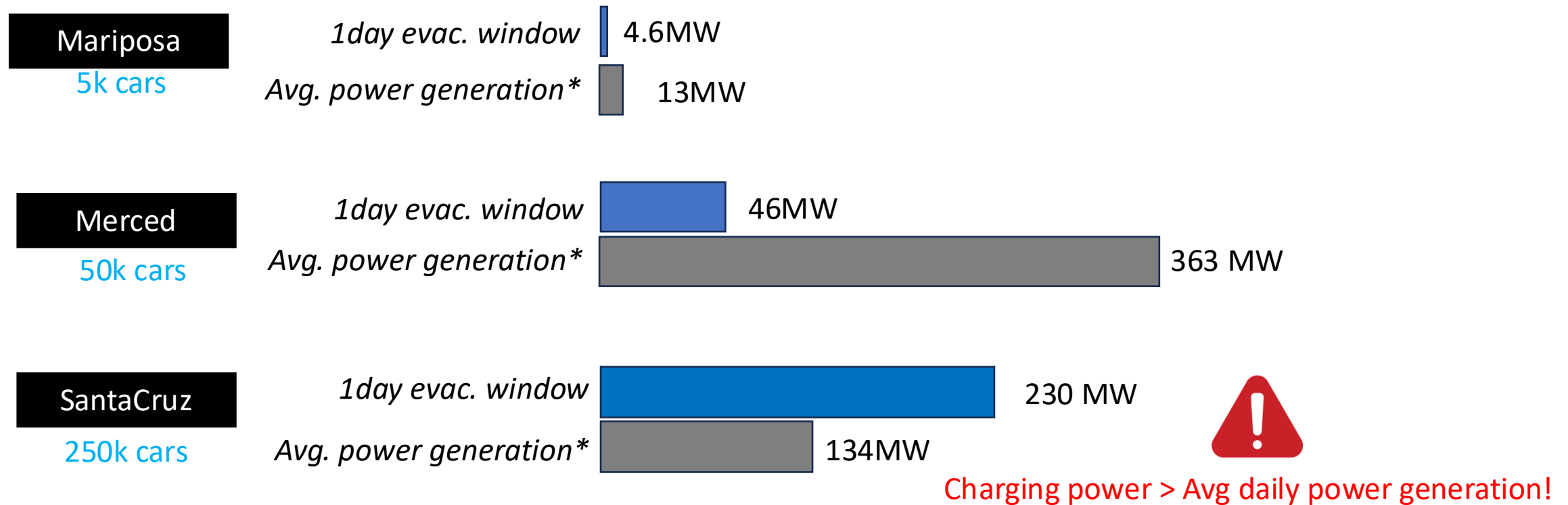
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Avg. Power Analysis



Challenge: Grid Overload



*What if we need to evacuate and the grid is **overloaded**?*